

Matan Melamed

Student for computer science at HUJI

CONTACT

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- Linkedin

SKILLS

- Proficient in programming languages : Java | C++ | C | Python | Assembly
- Experienced in : OOP | SQL | Algorithms | Data structures | System Programming

PROJECTS

Markov Node Linked List (C)

Developed a generic Markov Chain library in C with dynamic memory management and frequency-based transitions. Applied it to create a tweet generator and Snakes and Ladders simulator, emphasizing modular architecture, efficient data handling, and reusable components.

Bricker — Java (OOP, Strategy/Factory patterns)

Developed a modular Breakout-style game using advanced OOP principles. Implemented multiple brick behaviors through the Strategy and Factory design patterns, enabling features such as extra paddle, multi-ball, turbo speed, and life recovery. Designed a dynamic lives HUD and managed game flow with clear separation between rendering, logic, and event handling.

DNS Cache Poisoning — C (LDNS, Raw Sockets)

Developed a network security tool in C to execute a Kaminsky-style DNS cache poisoning attack against a BIND9 resolver. Implemented a distributed architecture consisting of a port-leaking server and an attacking client to capture and utilize resolver source ports via a TCP control channel. Utilized the LDNS library and raw sockets to craft and flood spoofed DNS responses, manipulating Authority and Additional sections to inject malicious records. Successfully demonstrated traffic redirection by winning Transaction ID (TXID) races and exploiting recursive lookup vulnerabilities within an isolated Docker environment.

LANGUAGES

- English (Fluent)
- Hebrew (Native)



PROFILE

Motivated and detail-oriented Computer Science student at the Hebrew University of Jerusalem, passionate about software development and algorithmic problem-solving. Skilled in multiple programming languages and experienced in building efficient, modular systems. Currently seeking a software development position where I can apply my technical and analytical skills in a practical environment.



EDUCATION

2022 - current

Hebrew University Of Jerusalem

Computer Science

Relevant Coursework: Data Structures, Algorithms, Object-Oriented Programming, Operating Systems, Databases, Cyber Security, Machine Learning

MILITARY SERVICE

Commander - Nativ Program, IDF

Led and mentored groups of soldiers in a preparatory education framework. Responsible for instruction, coordination, and personal development of participants, emphasizing leadership and teamwork.